Dr. Farhanul **Hasan**

Postdoctoral Fellow · Space Telescope Science Institute

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Education

New Mexico State University — *Las Cruces, NM*

Ph.D.: Astronomy; Degree received: Aug 2024; Advisor: Joseph Burchett — *Aug. 2018 - Aug. 2024*

Thesis: "On the Evolution of Galaxies in Cosmic Ecosystems"

M.Sc.: Astronomy; Degree received: Aug. 2022; Advisors: Joseph Burchett & Chris Churchill — *Aug. 2018 - Aug. 2022*

Reed College — *Portland, OR*

B.A.: Physics; Degree received: May 2018; Thesis advisors: Alison Crocker & Johnny Powell — *Aug. 2014 - May 2018*

Thesis: "Into the Void: Mass Function of Supermassive Black Holes in the local universe"

Experience

Space Telescope Science Institute — *Baltimore, MD*

Postdoctoral Research Fellow — *Sept 2024 - Present*

- Processing and analyzing grism spectroscopy and imaging data from James Webb and Hubble Space Telescopes
- Studying the physical properties of intermediate- and high-redshift galaxies, including metallicities, morphologies, and star formation rates
- Mapping the cosmic web, studying its gas properties and interplay with galaxy evolution, and making predictions for future galaxy surveys

New Mexico State University — *Las Cruces, NM*

Graduate Research Assistant — *Aug 2018 - Aug 2024*

- Processing and analyzing quasar spectroscopic observations and cosmological simulations
- Developing new tools for large-scale structure identification from simulated and observational data

Science Development Lead, *CosmoVis* — *May 2021 - Aug 2024*

- Developing and improving ***CosmoVis***, an open-source astrophysical 3D data visualization tool for exploring and analyzing hydrodynamical cosmological simulations of the universe, including identifying and implementing scientific uses and elucidating observations

Graduate Teaching Assistant — *Aug. 2018 - Dec. 2022*

Including ASTR 1115G/110G: Introduction to Astronomy, ASTR 305V: Life in the Universe, and ASTR 105G: The planets

Reed College — *Portland, OR*

Undergraduate Research Assistant — *May 2017 - Aug 2018*

Analyzing large SDSS photometric and spectroscopic catalogs and photometric images of nearby spiral galaxy NGC 4736

University of California, Irvine — *Irvine, CA*

Undergraduate Research Assistant — *June 2016 - Aug. 2017*

Analyzing space-based spectroscopic galaxy data, obtaining spectral energy distributions from UV to IR and measuring physical properties

Selected Publications

Hasan, F, et al., **"The Role of Star Formation and Morphology in the Mass-Metallicity Relation of Galaxies at Cosmic Noon"** submitted to *The Astrophysical Journal*, Oct. 2026. arXiv: 2610.37612.

Hasan, F, et al., **"Impact of Cosmic Filaments on Galaxy Morphological Evolution and Predictions of Early Cosmic Web Structure for Roman"** *The Astrophysical Journal*, Volume 1004, Number 1, Page 74, Jun. 2026.

Luber, N., **Hasan, F.**, et al. **"CHILES IX: Observational and Simulated HI Content and Star Formation of Blue Galaxies in Different Cosmic Web Environments"** *The Astrophysical Journal*, Volume 985, Number 2, Page 214, May 2025.

Hasan, F., et al., **"Filaments of The Slime Mold Cosmic Web And How They Affect Galaxy Evolution"** *The Astrophysical Journal*. Volume 970, Number 2, Page 177, Aug 2024.

Hasan, F., et al., **"The Evolving Effect Of Cosmic Web Environment On Galaxy Quenching"** *The Astrophysical Journal*, Volume 950, Number 2, Page 114, Jun 2023.

Hasan, F., et al., **"Evolution of CIV Absorbers. II. Where Does CIV Live?"** *The Astrophysical Journal*, Volume 924, Number 1, Page 12, Jan 2022.

Hasan, F., et al., **"Evolution of CIV Absorbers. I. The Cosmic Incidence"** *The Astrophysical Journal*, Volume 904, Number 1, Page 44, Nov 2020.

Acharyya, A.,..., **Hasan, F.**, et al., **"Spatially resolved gas-phase metallicity at $z \sim 2-3$ with JWST/NIRISS"**. *Astronomy & Astrophysics*. Volume 712. Number A136. Aug. 2026.

Goovaerts, I.,..., **Hasan, F.**, et al., **"MXDFz4.4: A LyC emitter 250 Myr after the epoch of reionization and a first test of Lyα morphology as a tracer of LyC escape at high redshift"**. *Astrophysical Journal*. Volume 1005, Number 1, Page 34. Jun. 2026.

Huberty, M..,..., **Hasan, F.**, et al., **"The NIRISS PASSAGE Spectroscopic Redshift Catalog in COSMOS"**. *Astrophysical Journal Supplements*. Volume 284, Issue 2, Page 64. Jun. 2026.

Malkan, M.,..., **Hasan, F.**, et al., **"Parallel Application of Slitless Spectroscopy to Analyze Galaxy Evolution (PASSAGE): Survey Overview "**. *The Astrophysical Journal*, Volume 993, Number 1, Page 152, Oct 2025.

Runnholm, A.,..., **Hasan, F.**, **"The JWST/PASSAGE Survey: Testing Reionization Histories with JWST's First Unbiased Survey for Lyman alpha Emitters at Redshifts 7.5-9.5"** *The Astrophysical Journal*, Volume 984, Number 1, Page 95, May 2025.

Nandi, G., Guo, Y., **Hasan, F.**, et al., **"Frozen in Size: Rest-frame Near-infrared Size Evolution of Quiescent and Star-forming Galaxies since Redshift $z = 4$ with JWST"**. in review, *Astrophysical Journal*. Apr. 2026.

Xu, P., Jiang, F., **Hasan, F.**, et al., **"Universal Dark-matter Density Profiles of Cosmic Filaments "**. in review, *Astrophysical Journal Letters*. Apr. 2026.

Finlator, K..,..., **Hasan, F.**, et al., **"Mapping the $z \geq 5$ SiIV Column Density Distribution onto the Galaxy Stellar Mass Function Using the Cumulative Absorption Cross Section"**. *Astrophysical Journal*. Volume 1008, Number 1, Page 62, Sept. 2026.

Woo, J., Cowan, M., **Hasan, F.**, et al., **"The Cosmic Web Origins of Blue-Centred Galaxies"**. in review, *Nature Astronomy*. Jul. 2025.

Awards & Grants

2025-	**PI: NASA High-End Computing Allocation SMD-24-54390923**, NASA Advanced Supercomputing Center	*Mountain View, CA*
2024-	**Science PI: Director's Discretionary Research Award**, Space Telescope Science Institute	*Baltimore, MD*
2024-	**Science PI: HST Cycle 31 Archival Research Award #17568**, Space Telescope Science Institute	*Baltimore, MD*
2023	**Arts & Sciences Travel Grant**, College of Arts & Sciences, New Mexico State University	*Las Cruces, NM*
2021-22	**Merit-based Enhancement Award**, Graduate School, New Mexico State University	*Las Cruces, NM*
2021	**Zia Award for Outstanding Research by a Graduate Student**, Dept. of Astronomy, NMSU	*Las Cruces, NM*
2019-20	**College of Arts & Sciences General Scholarship**, New Mexico State University	*Las Cruces, NM*
2018-20	**William Webber Voyager Fellowship**, Dept. of Astronomy, NMSU	*Las Cruces, NM*
2018	**Commendation for Academic Excellence**, Reed College	*Portland, OR*
2017	**Research Fellowship Award**, Reed College	*Portland, OR*
2017	**Research Opportunity Grant**, Reed College	*Portland, OR*
2016	**Summer Experience Award**, Reed College	*Portland, OR*

Selected Invited Talks

Filaments in the multi-scale Universe: understanding their role and detectability — *Kashiwa, Japan*

"Identifying cosmic filaments: novel advances, implications for galaxy evolution, and promise of wide-field surveys" — *Oct. 2026*

PASSAGE ISSI Collaboration Meeting — *Bern, Switzerland*

"Galaxy Morphologies and their Relationship with Mass, Star Formation, and Metallicity at Cosmic Noon" — *Feb. 2026*

University of Missouri, Columbia Astrophysics Seminar — *Columbia, MO*

"Galaxy Evolution in the Cosmic Web: from Large to Small Scales of Cosmic Ecosystems" — *Sep. 2025*

2025 Santa Cruz Galaxy Workshop — *Santa Cruz, CA*

"The Early Cosmic Web: Impact on Galactic Morphological Evolution and Predictions for Roman" — *Aug. 2025*

Baryon Pasting Collaboration 2025 meeting — *New Haven, CT*

"Modeling cosmic web filaments: challenges, advancements, and future directions" — *Apr. 2025*

Yale University Special Galaxy Seminar — *New Haven, CT*

"Cosmic Filaments: Friends of Foe of Galaxy Evolution?" — *Apr. 2025*

College University of New York (NYC College of Technology) Theoretical Physics Seminar — *New York, NY*

"The Slime Mold Cosmic Web and What it Teaches Us about Galaxy Evolution" — *Nov. 2024*

2024 Santa Cruz Galaxy Workshop — *Santa Cruz, CA*
"Filaments of the Slime Mold Cosmic Web" — *Aug. 2024*

Steward Observatory/NOIRLab Galaxy Group — *Tucson, AZ*
"A cosmic connection: How galaxy formation is affected by the cosmic web" — *Oct. 2023*

Columbia University Seminar — *New York, NY*
"A cosmic connection: How galaxy formation is affected by the cosmic web" — *Sep. 2023*

Yale University Galaxy Lunch — *New Haven, CT*
"A cosmic connection: How galaxy formation is affected by the cosmic web" — *Sep. 2023*

Arizona State University Borthakur Extragalactic Group — *Tempe, AZ (Virtual)*
"The Impact of Cosmic Web Structures on Galaxy Formation" — *Sep. 2022*

2022 Santa Cruz Galaxy Workshop — *Santa Cruz, CA*
"The Impact of Cosmic Web Structures on Galaxy Formation" — *Aug. 2022*

Selected Contributed Talks

Surveying the Universe in 4D: Beating Cosmic Variance with Wide-Field Slitless Spectroscopy — *Baltimore, MD*
"Revealing baryon cycling at the peak of galaxy formation with JWST/NIRISS WFSS " — *Aug. 2026*

Space Telescope Science Institute/Johns Hopkins University HotSci 2026 — *Baltimore, MD*
"Shaping Cosmic Ecosystems Through Baryon Cycling: Simulations, Observations, and the promise of the Roman Era" — *Aug. 2026*

The Baryon Cycle from Reionization to the Cosmic Noon — *Puerto Varas, Chile*
"The Evolving Role of the Cosmic Web in Growing and Shaping Galaxies" — *Dec. 2025*

"Some Like It Hot!" A Journey from the Hot IGrM to the Multiphase CGM — *Garching, Germany*
"A Cosmic Connection: The Impact of the Cosmic Web on Galaxy Evolution" — *May 2024*

243rd Meeting of the American Astronomical Society — *New Orleans, LA*
"A Cosmic Connection: The Effect of the Cosmic Web on Galaxy Evolution" (Dissertation Talk) — *Jan. 2024*

Oases in the Cosmic Desert: Understanding the Structure of the Circumgalactic Medium — *Tempe, AZ*
"The connection between galaxy quenching and cosmic web structures across cosmic time" — *Feb. 2023*

The Co-evolution of the Cosmic Web and Galaxies across Cosmic Time at KITP — *Santa Barbara, CA*
"How do cosmic web structures affect galaxy quenching across cosmic time?" — *Feb. 2023*

Service and Leadership

Organizer, "Low-Density Universe" Journal Club — *Baltimore, MD*
Space Telescope Science Institute — *Aug 2026 - Present*

Guest Lecturer, JHU pre-college "Exploring the universe with Space Telescopes" program — *Baltimore, MD*
Johns Hopkins University — *Jun 2026 – Aug 2026*

Local Organizing Committee, STScI Spring Symposium 2026 — *Baltimore, MD*
Space Telescope Science Institute — *Jan 2026 – Present*

Panel Support Scientist, JWST Cycle 5 Peer Review — *Baltimore, MD*
Space Telescope Science Institute — *Nov 2025 – Feb 2026*

HotSci 2025 Summer Colloquium Series Organizer — *Baltimore, MD*
Space Telescope Science Institute — *Apr 2025 – Aug 2025*

Postdoc Representative to the Science Mission Office — *Baltimore, MD*
Space Telescope Science Institute — *Feb 2025 – Present*

Journal Paper Referee — *Virtual*
ApJ, A&A, MNRAS, OJAA, RASTI — *May 2023 – Present*

Co-supervisor/Research Mentor — *Las Cruces, NM*
Department of Astronomy, New Mexico State University — *May 2022 - Aug. 2024*

Executive Secretary for NASA Astrophysics Proposal Review Panel — *NASA*
NASA ROSES — *May 2023*

President, Bangladesh Student Association (BSA) — *Las Cruces, NM*
New Mexico State University — *Aug. 2023 - Aug. 2024*

Mentor for Prospective Graduate Students — *Científico Latino*

Científico Latino Graduate Student Mentorship Initiative (GSMI) — *Jul. 2020 - May 2022*

Community Outreach Volunteer — *Las Cruces, NM*

Department of Astronomy, New Mexico State University — *Aug. 2018 - Aug. 2024*

Co Mentor, UC CROSS — *UC Santa Cruz*

Google Summer of Code 2022 — *Jun. 2022-Dec. 2022*